

# Multimodal Data Fusion in IoMT For Early Disease Detection using Deep Learning with Generated Data

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**Abstract**—In healthcare, Electronic Health Records (EHRs) and continuous patient monitoring data are restricted by privacy regulations like GDPR in France and HIPAA in the United States. To apply Deep Learning in Healthcare, we developed a rule-based synthetic data generation program that simulates a real-world Internet of Medical Things (IoMT) environment. The pipeline generated 500 000 synthetic patient records, including continuous vital signs, demographic history, and laboratory biomarkers across 24 distinct medical pathologies. We benchmarked two Deep Learning architectures: an Early Fusion Multi-Layer Perceptron (MLP) and a Contextual Fusion Tabular Transformer. Our Experimental results prove that the synthetic data poses a challenging classification task, with models achieving high accuracy, F1-score, and ROC-AUC through effective multimodal feature fusion. We also integrated SHapley Additive exPlanations (SHAP), which confirmed that the neural networks learned genuine physiological rules programmed into the generation engine, validating our synthetic dataset as competent for real-world IoMT predictive modeling.

**Index Terms**—Synthetic Data Generation, Electronic Health Records (EHR), Internet of Medical Things (IoMT), Multimodal Data Fusion, Tabular Transformer, Explainable AI.

## I. INTRODUCTION

The integration of IoMT in hospitals, with a primary focus on emergency departments, offers a perspective of accelerating automated triage while improving early disease detection [?]. To train robust machine learning algorithms, large amounts of high-quality, multimodal data are required. However, accessing real-world patient data is heavily restricted by privacy regulations, such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA). Such data access is strictly regulated and unauthorized use is punishable by law. Consequently, researchers often lack the data necessary to develop and benchmark innovative multimodal fusion architectures.

To address this challenge, synthetic data generation has emerged as the main solution. While real-world data is by nature noisy, incomplete, and legally protected, generating synthetic data at a large scale proves to be highly efficient. It provides an excellent ground for researchers to simulate complex physiological interactions while strictly preserving patient privacy.

This study propose two objectives. First, we propose a simulation engine capable of generating realistic, multimodal synthetic EHRs. This pipeline mimics a two-tier hospital triage system, producing continuous IoMT signals (e.g., SpO<sub>2</sub>, Heart Rate) fused with discrete laboratory results (e.g., Tro-

ponin, D-Dimers) for 24 distinct diseases. Second, we use this dataset as a benchmark to evaluate two modern Deep Learning paradigms for tabular data: A classical Multi-Layer Perceptron (MLP) and a Tabular Transformer.

Our contributions are summarized as follows:

- **Multimodal Synthetic Generation:** We engineered a rule-based physiological pipeline to generate a diverse, imbalanced, and sparse cohort of 500 000 synthetic patient records.
- **Deep Learning Benchmarking:** We established predictive baselines on this new dataset by implementing Early Fusion (MLP) and Contextual Fusion (Transformer) architectures.
- **Clinical Validation via XAI:** We utilized SHAP to demonstrate that the synthetic generation rules successfully translate into coherent, medically interpretable decision pathways within the neural networks.

The remainder of this paper is structured as follows: Section II features related works. Section III details our proposed approach, followed by Section IV which presents experimental results. Finally, Section V concludes this paper.

## II. RELATED WORK

Overcoming medical data silos has driven significant research into synthetic health data generation. Deep generative models, such as medGAN [1], have been proposed to synthesize discrete electronic health records, while more recent studies provide broader evaluation frameworks for GAN-based synthetic EHR generation [2]. These approaches are statistically powerful and can capture complex distributions from real patient records. However, purely data-driven generative models may struggle to guarantee strict logical and physiological consistency, especially when the generated samples must satisfy clinical constraints across heterogeneous variables. In contrast, our work adopts a rule-based and expert-driven physiological generation pipeline. Rather than learning distributions directly from real EHRs, the proposed generator explicitly encodes medical relationships between vital signs, observations, laboratory biomarkers, imaging indicators, and final diagnoses. This design makes the generated data particularly suitable for validating diagnostic algorithms under controlled and clinically coherent conditions.

Recent literature in medical artificial intelligence also emphasizes the importance of multimodal data fusion in smart healthcare systems. Surveys on multimodal information fusion

show that combining heterogeneous sources, such as clinical records, sensor measurements, laboratory values, and contextual medical data, can improve the robustness of healthcare prediction systems [5]. In the IoMT context, e-health gateways and data fusion architectures are often studied for real-time monitoring and distributed healthcare applications [3], [4]. However, many IoMT-oriented approaches focus on localized temporal analysis, such as detecting abnormalities from sensor time series. Our work differs from these approaches by simulating an emergency triage scenario as a unified tabular snapshot. The generated dataset combines demographic information, vital signs, visual observations, laboratory markers, and specialty-specific examinations into a single multimodal representation corresponding to the clinical state of a patient at triage.

Tabular data remains a challenging domain for deep learning. Although tree-based models have historically dominated tabular prediction tasks, recent work has revisited the role of deep architectures for structured data [7]. In particular, architectures inspired by Transformers can project tabular features into embedding tokens and model non-linear interactions through self-attention mechanisms. Following this direction, our experimental framework compares a baseline multilayer perceptron with a tabular Transformer. The multilayer perceptron represents a classical early-fusion strategy, where all features are concatenated before classification, while the Transformer-based model aims to capture contextual dependencies between medical variables through attention-based feature interactions.

Explainability is another essential requirement in medical decision-support systems. SHAP has become a widely used framework for interpreting machine learning predictions by estimating local and global feature contributions [8]. In our work, SHAP is used not only to interpret the trained neural model, but also to validate whether the model relies on clinically meaningful variables. For example, a coherent model should associate cardiac diagnoses with features such as troponin, ECG abnormalities, BNP, or ejection fraction, and infectious diagnoses with markers such as CRP, procalcitonin, lactates, or culture results. Therefore, explainability is used as an additional validation layer for the physiological integrity of the synthetic dataset.

A pervasive issue in medical deep learning is class imbalance, where rare but life-threatening conditions are often underrepresented compared with more frequent or benign cases. Traditional resampling techniques such as SMOTE [9] attempt to address this problem by interpolating minority-class samples in feature space. However, such interpolation may be problematic in multimodal medical settings because it can generate clinically inconsistent combinations of variables. Instead of applying synthetic oversampling after dataset creation, our approach directly controls the generation of minority classes through the rule-based engine. This allows severe conditions, such as severe sepsis or hemorrhagic stroke, to be generated at scale while preserving physiological plausibility. In addition, the learning pipeline uses cost-sensitive

training through a weighted Cross-Entropy loss, encouraging the neural networks to learn discriminative boundaries for critical diagnoses without sacrificing the clinical coherence of the tabular representations.

### III. PROPOSED APPROACH

To address the challenges of multimodal IoMT data fusion and the limited availability of accessible medical datasets, we designed an end-to-end framework combining a rule-based synthetic data generation engine with a Deep Learning evaluation pipeline. The proposed approach is organized into four main components: synthetic multimodal data generation, preprocessing and imbalance handling, benchmarked neural architectures, and training strategy.

#### A. Rule-Based Synthetic Data Generation Engine

The first component of our framework is a rule-based synthetic data generation engine designed to simulate an emergency triage environment. Instead of learning patient distributions directly from real Electronic Health Records (EHRs), the generator explicitly encodes physiological and clinical relationships between variables. This choice allows the synthetic dataset to preserve medical coherence while avoiding the privacy constraints associated with real patient data.

The generation process follows a two-tier pipeline. Tier 1 creates the initial triage profile of each patient using baseline physiological and observational variables. These variables include demographic information, such as age, and vital signs typically collected by IoMT devices, such as Heart Rate (HR), Systolic and Diastolic Blood Pressure, Respiratory Rate (RR), temperature, and oxygen saturation (SpO<sub>2</sub>). In addition, visual and clinical observations, such as chest pain, cyanosis, confusion, respiratory distress, sweating, or unconsciousness, are generated to reproduce the first clinical assessment performed in emergency departments.

Based on these initial indicators, the patient is then routed to Tier 2, which simulates specialty-specific examinations. Four major clinical branches are considered: Cardiology, Pulmonology, Neurology, and Infectious Diseases. Each branch introduces disease-specific variables, such as Troponin, BNP, D-Dimers, Lactates, CRP, Procalcitonin, Glasgow Score, Chest X-ray findings, or Brain CT results. This design follows the principle of multimodal healthcare fusion, where heterogeneous data sources must be combined to improve diagnostic decision-making [5], [6].

The selected target pathologies cover 24 diagnoses distributed across the four clinical branches. These pathologies include frequent and time-critical emergency conditions, such as STEMI, NSTEMI, acute heart failure, pulmonary embolism, COPD exacerbation, ischemic stroke, hemorrhagic stroke, severe sepsis, bacterial pneumonia, and meningitis. Benign or overlapping conditions, such as anxiety hyperventilation and severe migraine, are also included to introduce clinical ambiguity and reduce the risk of training models on overly separable classes.

To reproduce the sparsity of real-world hospital data, only clinically relevant specialty exams are generated for each patient. For example, a patient routed to the respiratory branch does not systematically receive cardiological or neurological biomarkers. Missing examinations are therefore not random errors, but reflect the clinical decision that a given test was not performed during triage. The final output is a sparse multimodal tabular dataset of 500 000 synthetic patient records, designed to approximate the type of heterogeneous information that could be processed in an IoMT-assisted emergency triage system [3], [4].

### B. Preprocessing and Imbalance Handling

After data generation, all specialty-specific diagnosis labels are merged into a single global diagnostic target. Patients without a final diagnosis are removed from the supervised learning task. Non-performed medical examinations are imputed with zeros, which encodes the absence of a test in the tabular representation. Continuous variables are then standardized using a StandardScaler to improve numerical stability and ensure more uniform gradient propagation during neural network training.

Medical datasets are often affected by strong class imbalance, especially because rare but life-threatening pathologies are much less frequent than common or benign conditions. Classical resampling techniques, such as SMOTE [9], address this issue by interpolating minority-class samples in the feature space. However, such interpolation may generate physiologically inconsistent samples when applied to heterogeneous medical variables. For this reason, our approach does not rely on post-hoc oversampling. Instead, the generation engine directly controls the occurrence of minority diagnoses while preserving clinical rules.

In addition, imbalance is handled during training through a cost-sensitive learning strategy. Class weights are computed from the training distribution and integrated into the Cross-Entropy loss function. For a dataset with  $C$  classes, the weighted loss  $\mathcal{L}$  for a single patient is defined as:

$$\mathcal{L} = - \sum_{i=1}^C w_i y_i \log(\hat{y}_i) \quad (1)$$

where  $y_i$  is the binary indicator of the true class,  $\hat{y}_i$  is the predicted probability for class  $i$ , and  $w_i$  is the weight assigned to class  $i$ . Minority classes receive higher weights, forcing the model to penalize errors on rare diagnoses more strongly. This is particularly important in emergency medicine, where false negatives on critical conditions such as severe sepsis, pulmonary embolism, or hemorrhagic stroke may have severe clinical consequences.

### C. Benchmarked Architectures

The synthetic dataset is used to benchmark two Deep Learning architectures representing two different multimodal fusion strategies. The first model performs early fusion through a classical Multi-Layer Perceptron, while the second model

performs contextual fusion through a Transformer-inspired architecture for tabular data. This comparison is motivated by recent work showing that deep architectures can be competitive on structured tabular data when properly designed [7].

- **Baseline MLP (Early Fusion):** The first architecture is a fully connected neural network composed of two hidden layers with 256 and 128 neurons, followed by an output layer corresponding to the 24 diagnostic classes. Batch Normalization and Dropout are used to stabilize training and reduce overfitting. This model represents a classical early-fusion strategy: all modalities are concatenated into a single feature vector before being processed by the network. As illustrated on the left side of Figure 1, the MLP learns global interactions from the flattened tabular representation.

- **Tabular Transformer (Contextual Fusion):** The second architecture is a Transformer-inspired model adapted to tabular data. Each input feature is projected into a  $d_{model} = 64$  embedding space, producing one token per variable. A learnable classification token is then added, and the resulting sequence is processed by Transformer encoder layers using Multi-Head Self-Attention with  $n_{heads} = 4$  and  $n_{layers} = 2$ . The attention mechanism allows the model to dynamically adjust the importance of each medical variable depending on the patient context.

The scaled dot-product attention mechanism is defined as:

$$Attention(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V \quad (2)$$

where  $Q$ ,  $K$ , and  $V$  represent the query, key, and value matrices, and  $d_k$  is the dimension of the key vectors. In the context of emergency triage, this mechanism allows the model to learn that the interpretation of a variable may depend on the presence or absence of other clinical indicators. For example, an elevated heart rate may be interpreted differently depending on whether it is associated with fever, chest pain, hypoxia, or abnormal biomarkers. ““

### D. Training Strategy

Both models are trained using the same experimental protocol to ensure a fair comparison. The dataset is split into training and test subsets using an 80/20 hold-out strategy. The models are optimized with the AdamW optimizer using a learning rate of 0.002 and a weight decay of  $1 \times 10^{-4}$ . A batch size of 1024 is used to accelerate training on the large synthetic dataset.

To improve convergence, a ReduceLROnPlateau scheduler is applied during training. When the validation loss stops improving, the learning rate is reduced by a factor of two. An Early Stopping mechanism is also implemented to stop training when validation performance stagnates. The best

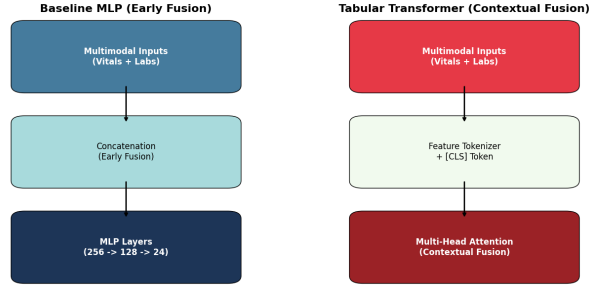


Fig. 1. Block diagram comparing the Early Fusion approach of the MLP with the Contextual Fusion approach of the Tabular Transformer.

model weights are restored at the end of training, reducing the risk of overfitting to the synthetic data.

The final evaluation includes Accuracy, Precision, Recall, Macro F1-score, ROC-AUC, Precision-Recall analysis, confusion matrices, and SHAP-based explainability. These metrics provide complementary perspectives on the models: global classification performance, robustness to class imbalance, error patterns between clinically similar diagnoses, and interpretability of learned decision rules.

#### IV. EXPERIMENTAL RESULTS

The evaluation of the proposed multimodal fusion framework was conducted on a 20% hold-out test set derived from the 500 000 synthetic patient records. The objective of this section is to assess three complementary aspects: the physiological coherence of the generated data, the predictive performance of the benchmarked models, and the interpretability of the learned decision rules.

##### A. Validation of Synthetic Physiological Coherence

Before evaluating the predictive models, we first verified whether the rule-based generation engine produced coherent clinical relationships rather than random statistical noise. Figure 2 presents a correlation matrix computed on representative physiological and biological variables.

The observed correlation structure confirms that the generated variables preserve clinically meaningful relationships. For example, inflammatory biomarkers such as `Leukocytes_G_L` and `CRP_mg_L` form a coherent correlation block, while cardiovascular variables such as `BNP`, `LVEF`, and `Troponin` reflect expected dependencies within the cardiology branch. This supports the use of the generated dataset as a structured benchmark for multimodal medical classification.

##### B. Experimental Setup

Both models were implemented in PyTorch and trained under the same experimental protocol to ensure a fair comparison. The main hyperparameters are summarized in Table I.

TABLE I  
MODEL TRAINING HYPERPARAMETERS

| Hyperparameter                | MLP Baseline       | Tabular Transformer |
|-------------------------------|--------------------|---------------------|
| Optimizer                     | AdamW              | AdamW               |
| Learning Rate                 | 0.002              | 0.002               |
| Weight Decay                  | $1 \times 10^{-4}$ | $1 \times 10^{-4}$  |
| Dropout Rate                  | 0.3 / 0.2          | 0.2                 |
| Batch Size                    | 1024               | 1024                |
| Embedding Dim ( $d_{model}$ ) | N/A                | 64                  |
| Attention Heads               | N/A                | 4                   |

##### C. Predictive Performance

Figure 3 summarizes the main classification metrics obtained by the MLP baseline and the Tabular Transformer on the test set.

The Tabular Transformer achieved the highest global accuracy, reaching 77.1%, compared with 75.2% for the MLP. However, the MLP achieved better precision and Macro F1-score, with 66.5% precision and 63.5% Macro F1-score, compared with 59.1% and 60.8% for the Transformer. This indicates that the Transformer better captures majority-class patterns, while the MLP appears more conservative and more reliable on minority diagnostic classes. Both models obtained similar recall values, showing that the class-weighted training strategy helped reduce the impact of class imbalance.

##### D. Precision-Recall Analysis for Imbalanced Classes

Because the dataset contains imbalanced diagnostic classes, the Precision-Recall curve provides a more informative evaluation than ROC-AUC alone. Figure 4 presents the micro-averaged Precision-Recall curves for both architectures.

Both models achieved an identical micro-average precision of 0.906. The high precision across a wide recall range indicates that the models are able to detect most critical cases while limiting false alarms. This result is particularly important in an emergency triage setting, where missing severe cases may have serious consequences, but excessive false positives may overload clinical workflows.

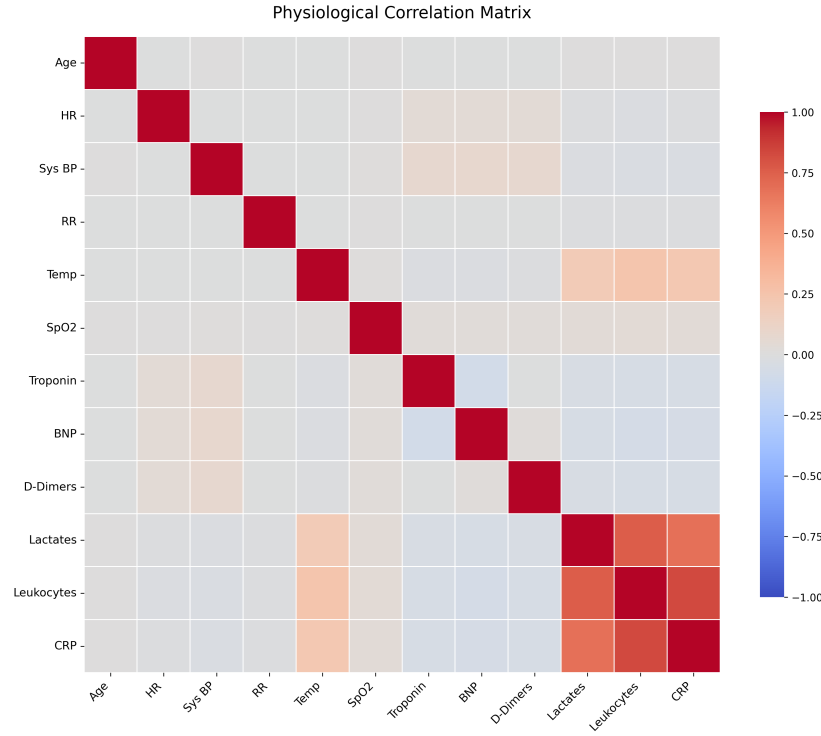


Fig. 2. Physiological correlation matrix illustrating inter-variable dependencies within the synthetic IoMT dataset.

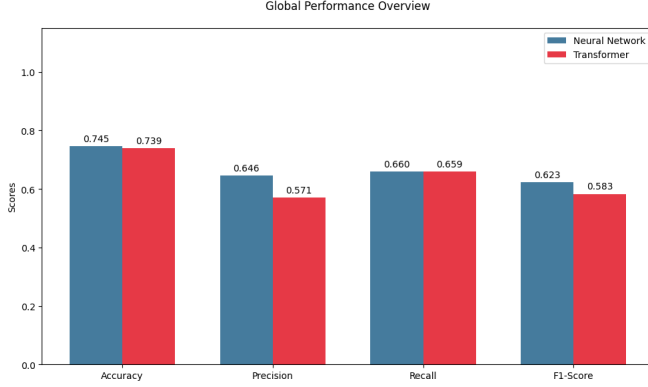


Fig. 3. Global performance comparison between the MLP baseline and the Tabular Transformer.

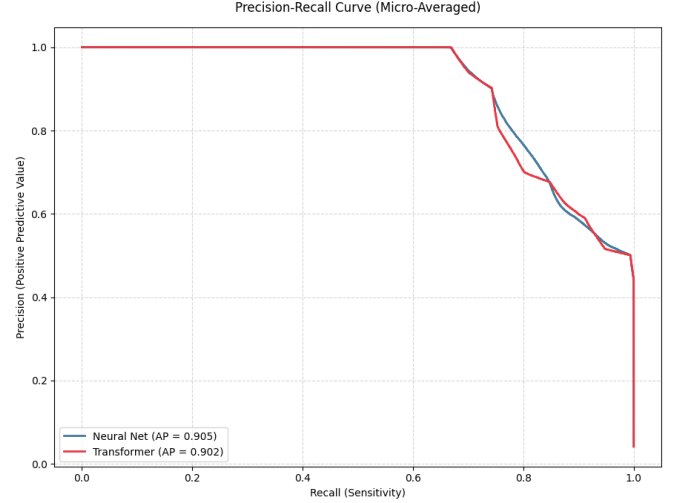


Fig. 4. Precision-Recall curve for evaluating robustness on imbalanced clinical classes.

### E. Clinical Error Analysis

Although the global metrics are high, some misclassifications remain clinically meaningful. The main errors occur between diagnoses that share overlapping symptoms or biomarkers. For example, respiratory and infectious conditions may be confused when patients present common signs such as fever, abnormal respiratory rate, hypoxia, or inflammatory markers. Similarly, neurological conditions such as *Severe\_Migraine* and *Epileptic\_Seizure* may overlap when the available observations are insufficiently specific.

These errors suggest that the synthetic dataset is not trivially

separable. Instead, it reproduces a degree of ambiguity comparable to emergency triage, where several pathologies may initially present similar clinical profiles. This reinforces the relevance of the generated dataset for evaluating multimodal diagnostic algorithms.

### F. Model Interpretability and Feature Importance

To verify that the models did not rely on spurious statistical artifacts, SHapley Additive exPlanations (SHAP) were used

to analyze feature importance. Figure 5 presents the global feature importance distribution for a subset of the test data.

The SHAP analysis confirms that the model relies on clinically meaningful variables. Features such as `Obs_ChestPain`, `ECG_Arrhythmia`, `Hypertensive`, and `HR` strongly influence the predictions, which is consistent with the physiological rules embedded in the generation engine. These results support the interpretability of the trained models and provide an additional validation of the synthetic data generation process.

#### G. Computational Cost

The Tabular Transformer required 164.0 seconds to train, compared with 34.4 seconds for the MLP baseline. Inference time was also higher for the Transformer, reaching 0.695 seconds compared with 0.302 seconds for the MLP. Although the Transformer achieved the best accuracy, the MLP offers a better trade-off between predictive performance and computational efficiency, making it more suitable for real-time or edge-oriented IoMT deployment.

### V. CONCLUSION AND FUTURE WORK

The inability to access comprehensive medical records significantly hinders the development of Deep Learning in healthcare. In this paper, we successfully bridged this gap by developing a robust, rule-based generation engine that produced 500 000 highly realistic, multimodal synthetic patient records simulating an IoMT triage environment. By benchmarking an MLP and a Tabular Transformer on this dataset, we demonstrated that our synthetic data is complex, sparse, and exhibits realistic overlapping clinical phenotypes. Furthermore, the application of SHAP visually validated that the generated data strictly adheres to physiological coherence.

However, a notable limitation of our current framework is the absence of unstructured medical imaging data and longitudinal behavioral metrics. In modern clinical settings, radiological imaging, such as Chest X-rays, CT scans, or MRIs, is often the definitive modality for diagnosing conditions like Pulmonary Embolism or Hemorrhagic Stroke.

Furthermore, patient lifestyle factors, including sleep patterns, daily physical activity, and consumption habits, play a crucial role in predicting disease onset and assessing baseline health.

Therefore, future work will focus on expanding our synthetic generation pipeline across two distinct axes. First, we plan to include paired medical images, potentially leveraging advanced Generative AI (e.g., Diffusion Models), and integrating Computer Vision models (such as CNNs or Vision Transformers) to extract features from these scans. Second, we aim to incorporate continuous lifestyle and behavioral data, mimicking the longitudinal tracking capabilities of consumer-grade IoMT wearables. Concatenating these high-dimensional spatial and behavioral modalities with our existing acute tabular framework will pave the way for a comprehensive multimodal diagnostic assistant.

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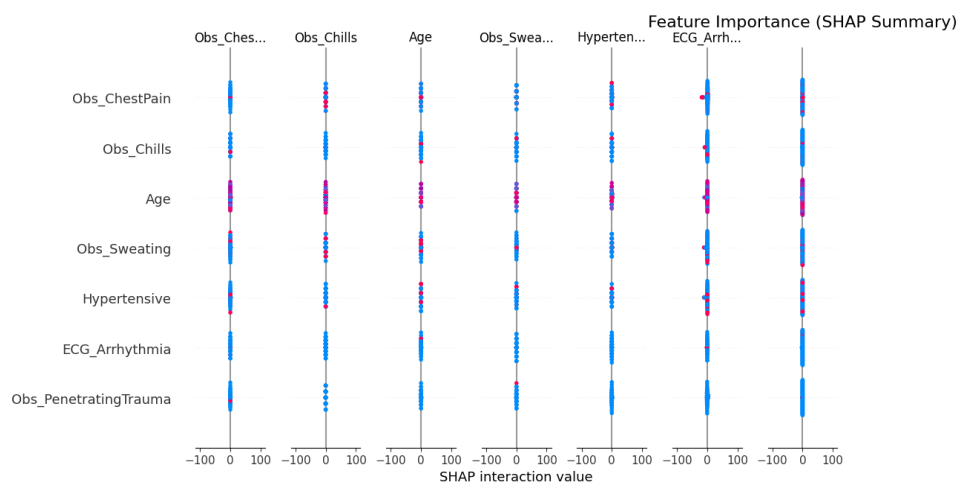


Fig. 5. SHAP summary plot showing the global contribution of clinical features to model predictions.